FACULTY OF SCIENCE

UNIVERSITI TUNKU ABDUL RAHMAN

PROJECT REPORT GUIDE

CHAPTER 1

GENERAL REQUIREMENT OF PROJECT REPORT

1.1 Introduction

The aim of this guide is to help students of Universiti Tunku Abdul Rahman in the preparation of their final year project report. Students are advised to acquaint themselves with the recommendations of this guide in the early stages of preparation, even though some of the recommendations apply only to the final stages. The Guide also provides information for supervisors whose responsibilities include ensuring that the students follow its recommendations.

- **A project report which does not follow the approved format will not be accepted and the degree will not be conferred until an acceptable standard has been met.**

1.2 Project Report Preparation

The project report may be written in **British** or **American English**. However, for consistency, the candidate **must use only one spelling system** and not a mixture of both. The project report must form a distinct contribution to the knowledge of the subject and afford evidence of originality. It must be of a satisfactory literary standard and must be suitable for publication as a project

report of Universiti Tunku Abdul Rahman.

The project report must consist of the candidate's account of his/her own research. A candidate will not be permitted to submit a project report consisting wholly of work for which a degree has been conferred on him/her in this or any other university.

A project report must be submitted in a form that can be reproduced in a clear and usable format.

Close attention should be paid to the following criteria:

i. The text and all illustrative materials should be clear and error free;

ii. Good quality paper should be used;

iii. Margins on each page should be as specified in this Guide.

Two soft bound copies must be submitted to the Department for examination. The report, including tables and figures, must not exceed 100 pages (excluding Appendices) and both copies should be bound with soft covers. Upon examination both the soft bound copies will be returned to the

candidate. **After making the necessary corrections, the candidate is required to submit ONE softcopy of the project report in a CD (in PDF format) and a hardcopy of the Declaration form (Appendix D) and Approval (Appendix E) and Permission sheets (Appendix F) to the university; and a CD and/or hardcopy to the respective supervisor(s).** An abbreviated version of the degree, your name and the year should be printed on the stem of the hard bound copy of your project report.

1.3 Technical specifications

1.3.1 Collation

It is the candidate's responsibility to prepare and assemble all materials for the project report, and to have the pages of the project report in correct order.

1.3.2 Paper

The **project report must be written on one side only** of good quality white bond paper (usually of 80g weight) of **A4 size (210mm x 297mm)**. The same grade of paper must be used throughout the project report.

1.3.3 Pagination

All pages should be numbered right flushed at the bottom margin. Page numbers should appear by themselves and should not be placed in brackets, hyphenated or accompanied by other decorative devices. Print text or figures on only one side of each sheet. Only original word-processed copy or good and clean photocopies are acceptable.

Pages should be numbered consecutively throughout the project report, including pages for tables, figures and appendices. Each appendix should be identified separately using an uppercase letter beginning with **A**. The pages of the appendices should also be numbered accordingly.

The Preliminary pages preceding Chapter 1 must be numbered in Roman numerals (i, ii, iii,…). The Title Page should not be numbered though it is counted as page i. Page 1 begins with the first page of the Introduction (or Chapter 1) but not numbered. Subsequent pages should be numbered beginning with page 2. Each chapter should begin on a new page.

1.3.4 Production of Manuscript

Project reports must be produced on a word processor with printing done on a high quality printer.

1.3.5 Spacing

Double line spacing must be used in between lines. However, four line spacing must be used between paragraphs and sections. **Single spacing is permitted within**:

i. Explanatory footnotes

ii. Quotations longer than three lines set in a block

iii. References (except between entries)

iv. Multi-line captions (tables, figures)

v. Appendices

vi. Headings or subheadings for Tables and Figures

A new paragraph should be at least two lines at the bottom of the page. Otherwise, the new paragraph should start in a new page.

1.3.6 Margins

The top, bottom and right margins should be not less than 1" or 2.5 cm in width,

and the left-hand margin should be 2" or 5 cm in width to allow for loss in binding. Right margin of the text should be justified.

1.3.7 Typeface and Font Size

The **font size should be 12 point**, and the same typeface must be used throughout the text, including heading and page numbers. Font should not be scripted or italicized except for scientific names and terms in a different language. **Bold print may be used for headings**. Footnotes and text in tables should not be less than 8 point. Equations and formulae must be typed.

Some appropriate fonts for project report are:

i. Arial

ii. Times New Roman

1.3.8 Correction of Errors

The project report must be free of typographical errors. Corrections made in ink or with opaque fluid are not acceptable. If a neat erasure cannot be made the page must be re-typed. Deletions must not be made by crossing out or striking out letters or words. The typed line must be continuous. If the word or words

cannot be erased clearly and neatly with proper spacing, the whole line, the paragraph, or the whole page must be re-typed.

Additions of words or phrases must not be typed between lines or inserted by hand. The whole page must be re-typed to incorporate the additions.

Pages which are obviously insert pages are not acceptable. The surrounding pages should be re-typed to incorporate the material.

1.3.9 Binding

For candidates who wish to present a hardcopy of the project report to their supervisor(s), the **report should be bound in hard cover [unless otherwise requested by the supervisor(s)]**. The binding should be as for a book in which pages are permanently secured.

The following should be lettered in Gold with 18 point font size from the head to the foot of the project report **spine** (Refer to Appendix A)

i. Name of candidate

ii. Abbreviated degree (e.g., BSc. (Hons) Biotech.)

iii. Year of submission

The following particulars should be provided on the project report **cover** using **Gold** block font with font size 18 point. Refer to Appendix A:

i. Title of project report

ii. Name of candidate

iii. Degree

iv. Name of University

v. Year of submission

CHAPTER 2

FORMAT OF PROJECT REPORT

Generally, the project report consists of three major parts: preliminary pages, text or main body and ending pages.

A) Preliminary Pages

Include the title page, dedication (optional), abstract, acknowledgement, declaration sheet, approval sheet, permission sheet, table of contents, list of tables, figures, and abbreviations.

B) Text or Main Body

Divided into chapters and sections.

C) Ending Pages

Consist of references and appendices.

Below is the arrangement of various pages in a project report:

Items	Remarks
Title Page	Not to be paginated but counted as i. Subsequent pages are paginated and are numbered consecutively.
	Appendix B
Dedication (if any)	
Abstract	Appendix C
Acknowledgement	
Declaration	Appendix D
Approval Sheet	Appendix E
Permission Sheet	Appendix F
Table of Contents	Appendix G
List of Table(s)	Appendix H
List of Figure(s)	Appendix H
List of Abbreviations	Appendix K
Main Body (Chapters)	
References	Appendix L
Appendices	

2.1 Title Page

Title should describe the content of the project report accurately and concisely.

The title page should include:

i. The title page should look exactly like the sample on Appendix B.

ii. Full title of project report (uppercase)

iii. Full name of author (uppercase)

iv. Name of institution to which the project report is submitted

v. Name of faculty

vi. Degree of which the project report is submitted

vii. Month and year of submission

viii. This page is counted but not numbered (it counts as page i but should not have a page number).

2.2 Abstract

An abstract is a concise summary of the project report, intended to inform prospective readers about its content. It usually includes a brief description of the research, the procedures or methods, and the results and conclusions. An abstract should not include internal headings, parenthetical citations of items listed in the reference section, diagrams, or other illustrations. The abstract

should be in **one paragraph** and **less than 300 words**. In short, an abstract actually points out what has been done, what was found and what can be concluded.

Remember: an effective abstract gives information about the quality of the research as well as the investigator.

An example of abstract is shown in Appendix C.

2.3 Acknowledgement

Acknowledgement is used to express the author's appreciation for guidance and assistance from individual(s) and institutions.

2.4 Table of Contents

Contains the list of all the chapter titles in **upper case**, preceded by their numbers in Arabic numerals. Subheadings should be in **title case** (Refer to Appendix G).

Not more than three levels of subheadings are allowed in the Table of Contents (chapter names are level 1 headings, subheadings of a chapter are level 2 headings, subheadings of subheadings are level 3). Subsections of chapters may be numbered with Arabic numerals.

i. The Table of Contents should follow closely the sample in Appendix G.

ii. The heading **Table of Contents** must appear at the top of the first page. The heading must be bold, centered, and without punctuation.

iii. List chapter titles and at least the first or second order subdivisions. Make sure that chapter and section titles are worded exactly as they appear in the body of the project report.

iv. Single space each entry, but double space between entries.

v. Separate titles from page numbers with right-justified tabs and dot leaders. **Do not use periods to separate titles and page numbers.**

vi. These pages are counted and numbered in lowercase Roman numerals.

2.5 List of Tables

Project reports that contain tables should provide a List of Tables (Appendix H). The list should show the exact title of captions of all tables in the text, together with the beginning page number of each table. Tables should be numbered consecutively within each chapter.

The following guidelines apply:

i. The table number and accompanying title is placed **above** the body of the table and left justified (Appendix I).

ii. Table titles should be focused, descriptive, and short.

iii. Use a solid line above and below the body of the table, thus delimiting the table.

iv. Use horizontal lines within the heading sections to encompass several common elements.

v. Excessively large and complex tables that require multiple pages are discouraged, but if necessary, secondary pages should state "Table X continued" and repeat the headings.

vi. Table footers are used to explain symbols, give notes on a statistical analysis, define abbreviations, amplify certain aspects of the table, or recognize sources.

2.6 List of Figures

Project reports that contain figures should include a List of Figures (Appendix H). The list of figures includes graphs, photographs, charts and printed images. Figures should be numbered consecutively within each chapter.

The following guidelines apply:

i. Figure captions should be placed immediately **below** the figure. This includes the figure number (e.g., Figure 1.1) followed by a concise, descriptive statement. The explanatory text should always precede the figure (Appendix J).

ii. Colour can be used in graphs, maps, photos, etc.

iii. Glossy prints, photomicrographs, or other figures not produced on the regular project report paper should be mounted.

Figures, tables, and illustrations must be self explanatory. Generally, a figure, table, or illustration will occupy a separate page of the project report. However, if it is integrated within a page of text, there must be sufficient space before and after to clearly set it apart from the text. The text should cite a figure or table when those data are noted. For example: “Graduate enrolment at Missouri State University has increased during each of the last five years (Figure 1.1).”

2.7 List of Abbreviations

When many abbreviations and/or symbols are used, a List of Abbreviations with appropriate definitions should be included for the sake of clarity. Such lists

should be included following the List of Tables and List of Figures. When abbreviations or other symbols are used, they should be uniform and consistent throughout the project report. This list serves as ready reference to readers not familiar with the abbreviation. The list should be sorted in alphabetical order. Universally recognized scientific symbols (cm, mm, kg, etc.) need not be listed. Appendix K shows the example.

2.8 Text of Project Report

The major parts of a project report may contain such subjects as introduction, general argument, description of experimental procedures, research methods, discussion, summary, conclusions, recommendations, and other pertinent topics that are necessary for a complete understanding on the part of a sophisticated or professional reader.

i. The chapter and section titles should be the same as the headings listed in the Table of Contents. The pages are counted and numbered in Arabic numerals. The text begins four lines below this heading.

ii. The text must be double spaced. (Note that the heading lines are single spaced.)

iii. The pages are counted and numbered in Arabic numerals.

iv. First line of each paragraph should not be indented.

2.9 Headings in Text

Chapter headings must be bold and centered.

The chapter number and title must be separated and double-spaced as shown in the example below:

CHAPTER 1

INTRODUCTION

Headings must not be larger than 12 points.

Headings and subheadings must be double spaced.

2.10 Introduction

Introduction is the introductory chapter (background information) that indicates the problem to be addressed and its importance and validity. It sets forth the context, the hypotheses to be tested and the research objectives to be attained.

2.11 Literature Review

A **literature review** surveys scholarly articles, books and other sources (e.g., dissertations, conference proceedings) relevant to a particular issue, area of research, or theory, providing a description, summary, and critical evaluation of each work. The purpose is to offer an overview of significant literature published on a topic. However, a literature review is not a summary of works of different authors. Justification is given in this section for the use of specific solution techniques or problem solving procedures in your work.

2.12 Materials and Methods

Descriptions of materials, equipment, procedures, techniques, calculations and calibration plots employed through out the study. Limitations, assumptions and range of validity of the methods are also included in this section. This information is required to allow the reader to assess the believability of your results, and also is needed by another researcher to replicate your experiment.

2.13 Results

The results are actual statements of observations, whether positive or negative.

It can be reported in the form of figures, tables, statistics or text. Sufficient details should be presented so that others can draw their own inferences and construct their own explanations. You can break up your results into logical segments by using subheadings.

2.14 Discussion

A discussion is based on the results obtained from the study in relation to the hypotheses and is not simply a restatement of the results. It highlights the main finding, their significance and implications. The discussion section should be a brief essay in itself, answering the following questions:

- What are the major patterns in the observations?
- What are the relationships, trends and generalizations among the results?
- What are the likely causes (mechanisms) underlying these patterns?
- Is there agreement or disagreement with previous work?
- How do the obtained results relate to your hypothesis/problem statement in the introduction?
- Include possible explanations related to your results (must be supported by references).
- What are the contributions of your findings to your field of study?

Outline and discuss also the **future research work** as a separate subsection in the Discussion. The future work can arise from the results that you generated or work that you would like to undertake to confirm your hypothesis.

2.15 Conclusions

A summary which highlights the most important observations and outcomes of the study upon which a conclusion is drawn in line with the objective set. It also outlines the significance and contribution of your work to the area your of study.

2.16 Chapter Layout

Text in a chapter can be written into major sections and subsections. Major section is numbered as first level (e.g., 1, 2, 3,….) and subsection is numbered second level, and etc (e.g., 1.1, 1.1.1, 1.2.3,…..). Candidates should ensure that it is consistent throughout the project report and should be limited to a maximum of 4 levels.

2.17 References

The project report should contain appropriate documentation, i.e., references to

original literature relevant to the research presented in the project report.

The format used by UTAR for in-text literature citations and the list of references is based on the **Harvard Referencing System**. Students must adhere to the in-text citation and referencing styles of the Harvard Referencing System prepared by the Universiti Tunku Abdul Rahman Library (Appendix L).

The List of References should follow closely the sample in Appendix L.

i. The heading **References** must appear at the top of the first page. The heading must be bold, centered, and without punctuation.

ii. Begin the list of sources four spaces below the heading.

iii. Single space within each entry and double space between entries.

iv. Do not split entries over two pages. If an entire entry does not fit on one page, place the entire entry on the next page.

v. Entries are arranged according to alphabetical order.

vi. These pages are counted and numbered in Arabic numerals.

Every reference cited in the text (with the exception of personal

communications) should be listed in the List of References section that follows the main body of the project report. Likewise, citation which is not mentioned in the text should not be listed in the reference section. This one-to-one correspondence between citations and listing of references is essential.

Citations. The citations in the text are done using the Harvard Referencing System. Candidates should adhere closely to the examples in Appendix L.

2.18 Appendices

If supplementary original data, illustrative material, a detailed derivation of equations, an extensive proof of a theorem, or a quotation too extensive for the body of the project report are included, they may be presented in appendices. Similar material should be gathered in a single appendix.

i. Number the appendices consecutively in alphabetical order (**Appendix A**, **Appendix B...**).

ii. The heading **Appendix A** (or **B, C**) and the title of the material must be bold, centered, and without punctuation.

iii. The appendix material should begin four spaces below the heading.

iv. The pages are counted and numbered in Arabic numerals.

CHAPTER 3

WRITING CONVENTION

3.1 Units of measure

Arabic numeral should be used before measurement units and should not be spelt out, e.g., 5 kg, not five kg. Exemption is given to the first word of the sentence.

3.2 Numbers

All numbers should be spelt out unless they are followed by units of measure. Arabic numeral should be used for number 10 or more than 10. If a sentence begins with a number, the number should be spelt out. e.g., “Three replicates were done for the following experiment’.

Appendix A

Example of Spine and Cover of the Project Report

YVONNE CAROLINE **B.Sc. (Hons) Biotechnology** **20xx**	**CADMIUM BIOSORPTION USING FREE AND IMMOBILIZED BIOASS OF *ASPERGILLUS AWAMORI*** **YVONNE CAROLINE** **BACHELOR OF SCIENCE (HONS) AGRICULTURAL SCIENCE / BIOTECHNOLOGY / CHEMISTRY / BIOCHEMISTRY / BIOMEDICAL SCIENCE / FOOD SCIENCE / MICROBIOLOGY / STATISTICAL COMPUTING AND OPERATIONS RESEARCH / LOGISTICS AND INTERNATIONAL SHIPPING** **FACULTY OF SCIENCE** **UNIVERSITI TUNKU ABDUL RAHMAN** **MAY / OCTOBER 20xx**

Appendix B

Example of Title page

CADMIUM BIOSORPTION USING FREE AND IMMOBILIZED BIOMASS OF *Aspergillus awamori*

By

YVONNE CAROLINE

A project report submitted to the Department of XXXXX
Faculty of Science
Universiti Tunku Abdul Rahman
in partial fulfilment of the requirements for the degree of
Bachelor of Science (Hons) Agricultural Science / Biotechnology / Chemistry / Biochemistry / Biomedical Science / Food Science / Microbiology / Statistical Computing and Operations Research / Logistics and International Shipping

May / October 20xx

Appendix C

Format of Abstract

ABSTRACT

CADMIUM BIOSORPTION USING FREE AND IMMOBILIZED BIOMASS OF *Aspergillus awamori*

Yvonne Caroline

The ability of free and immobilized cells of *Aspergillus awamori* to act as an effective cadmium biosorbent was investigated using……From the study, …… However, ……..

Appendix D

Declaration

DECLARATION

I hereby declare that this final year project report is based on my original work except for quotations and citations which have been duly acknowledged. I also declare that it has not been previously or concurrently submitted for any other degree at UTAR or other institutions.

_____

Name

Appendix E

Example of Approval sheet

APPROVAL SHEET

This final year project report entitled "**CADMIUM BIOSORPTION USING FREE AND IMMOBILIZED BIOMASS OF *Aspergillus awamori*"** was prepared by YVONNE CAROLINE and submitted as partial fulfilment of the requirements for the degree of Bachelor of Science (Hons) Agricultural Science / Biotechnology / Chemistry / Biochemistry / Biomedical Science / Food Science / Microbiology / Statistical Computing and Operations Research / Logistics and International Shipping at Universiti Tunku Abdul Rahman.

Approved by:

_____

(Prof./Assoc. Prof./Dr./Mr./Ms. ABCDEF) Date:……..
Supervisor
Department of XXXXX
Faculty of Science
Universiti Tunku Abdul Rahman

Appendix F

Example of Permission Sheet

FACULTY OF SCIENCE
UNIVERSITI TUNKU ABDUL RAHMAN

Date: _____

PERMISSION SHEET

It is hereby certified that **YVONNE CAROLINE** (ID No: **04UEB12345**) has completed this final year project report / dissertation / thesis* entitled "THE EFFECTS OF MOODS IN ATTITUDE JUDGMENTS" under the supervision of _____ (Supervisor) from the Department of _____, Faculty of Science, and _____ (Co-Supervisor)* from the Department of _____, Faculty of Science.
(2 spaces)

I hereby give permission to the University to upload the softcopy of my final year project report / dissertation / thesis* in pdf format into the UTAR Institutional Repository, which may be made accessible to the UTAR community and public.

Yours truly,

_____
(YVONNE CAROLINE)

*Delete whichever not applicable

Appendix G

Example of Table of Contents

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ACKNOWLEDGEMENTS			**iv**
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Example of List of Tables / List of Figures

LIST OF TABLES / LIST OF FIGURES

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Appendix I

Example of Table Layout

Table 1.1: Graduate school enrolment at Universiti Tunku Abdul Rahman.

	Fall Semester Headcount [1]		
Year	Male	Female	Total
1996	640	1415	2055
1997	734	1462	2196
1998	935	1792	2727
1999	991	1882	2873
2000	1052	1952	3004
2001	1080	2025	3105
2002	1095	2175	3270

[1] Includes full-time and part-time students.

Appendix J

Example of Figure Layout

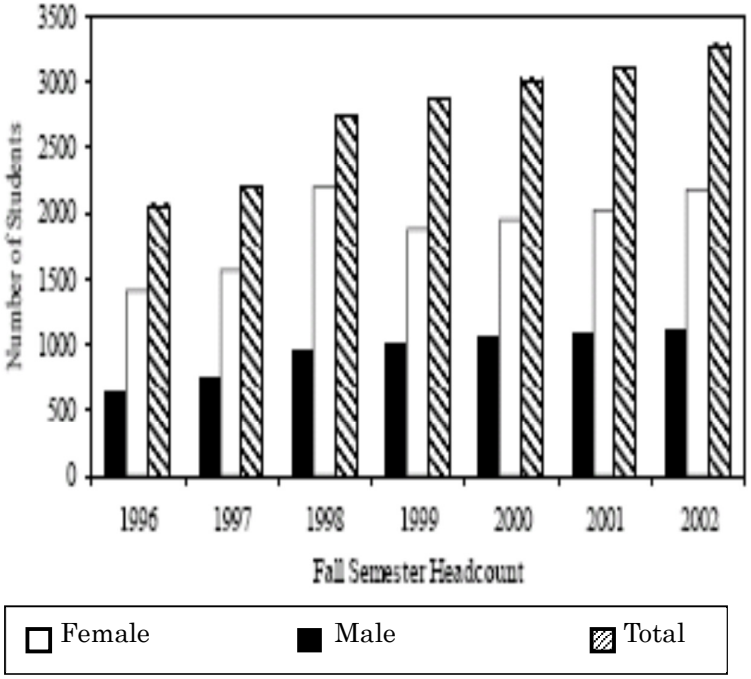


Figure 1.1: Graduate school enrolment at Universiti Tunku Abdul Rahman.

Appendix K

Example of List of Abbreviations

LIST OF ABBREVIATIONS

g	Acceleration of gravity (~9.8 m/s^2)
α-D-glucopyranose	Alpha-D-glucopyranose
BSA	Bovine serum albumin
Ca^{2+}	Calcium ion
C	Carbon
C/N ratio	Carbon-to-nitrogen ratio
Co^{2+}	Cobalt ion
CFU	Colony forming units
CD	Cyclodextrin
DP	Degree of polymerization
DEAE-cellulose	Diethylaminoathyl-cellulose
DOT	Dissolved oxygen tension
EC	Enzyme commission number

Appendix L

Harvard Referencing System

Please refer to ***pages 42-43*** *of the* ***UTAR Thesis/Dissertation Guidelines GD-IPSR-PSU-001 Rev. No. 4*** *(approved by the 214th Senate Meeting on 06/08/2013):*
http://research.utar.edu.my/ipsrweb/Postgraduate%20Handbook/guidelines/UTAR_Thesis_Dissertation_Guidelines%20%28Rev4%29%20%28214th%20Senate%2006.08.13%29%20Clean%20copy%20%28Amended%2012.11.13%29.pdf

Appendix P

Examples of Reference Format (Harvard style)

Journal:

Ahmad, F.B. and William, P., 1998. Rheological properties of sago starch. *Journal of Agriculture and Food Chemistry*, 46 (1), pp. 4060 – 4065.

Bakshi, A., Patnaik, P.R. and Gupta, J.K., 1992a. Pullulanase and α-amylase production by a *Bacillus cereus* isolate. *Letters in Applied Microbiology*, 14, pp. 210 – 213.

Bakshi, A., Patnaik, P. and Gupta, J.K., 1992b. Thermostable pullulanase from a mesophilic *Bacillus cereus* isolate and its mutant UV 7.4. *Biotechnology Letters*, 14, pp. 689 – 694.

Kubo, A. et al., 1999. The starch-debranching enzymes isoamylase and pullulanase are both involved in amylopectin biosynthesis in rice endosperm. *Plant Physiology*, 121(2), pp. 399 – 410.
* more than 5 authors

Conference paper (published):

Peter, S., 2009. Effects of child care on young children. *Proceedings of the third annual meeting of the International Society for Child Psychology*, 3 -5 December 2009 Atlanta, Georgia. Atlanta: Wiley, pp. 44 - 46.

Book:

John, Y.A. and Tim, M.A., 2009. *Thermodynamics: an engineering approach,* 2nd ed. London: McGraw Hill.

Book Chapter:

Bigelis, R., 1993. Carbohydrases. In: Nagodawithana, T., and Reed, G. (eds.). *Enzymes in food processing*. New York: Academic Press Inc., pp. 121 – 147.

Busche, R.M., 1985. Acetic acid manufacture – fermentation alternati*ves*. In: Cheremisinoff, P. N., and Ouellette, R.P. (eds.). *Biotechnology: applications and research*. Lancaster: Technomic Publishing Company, pp. 88 – 102.

Fogarty, W.M., and Kelly, C.T., 1990. Recent advance in microbial amylases. In: *Microbial Enzymes and Biotechnology*. 2nd ed. London and New York: Elsevier Applied Science, pp. 71 – 132.

Forst, G.M., 1986. Commercial production of enzymes. In: Hudson, B.J.F. (ed.). *Developments in food protein – 4*. London: Elsevier Applied Science Publishers, pp. 57 – 81.

Maloy, S.R., Cronal, J.E., and Freifelder, D., 1994. Mutagenesis, mutations and mutants. In: *Microbial Genetic*, Massachusetts: John and Bartlett Publishers, pp. 179 - 208.

Webpages:

Chaplin, M., 2004, *Production of glucose syrup* [Online]. Available at: http://www.lsbu.ac.uk/biology/enztech/glucose.html [Accessed: 1 March 2005].

Hedley, C., 2002, *Starch structure* [Online]. Available at: http://www.jic.bbsrc.ac.uk/staff/cliff-hedley/starch.htm [Accessed: 1 March 2005]

Thesis:

Kamariah, L., 1997. *Production, properties and applications of mycelium-bound lipase of a locally isolated strain of Aspergillus flavus link*. PhD thesis, Universiti Putra Malaysia, Malaysia.

Hobel, C.F.V., 2004, *Access to biodiversity and new genes from thermophiles by special enrichment methods.* PhD Thesis, University of Iceland, Reykjavik.